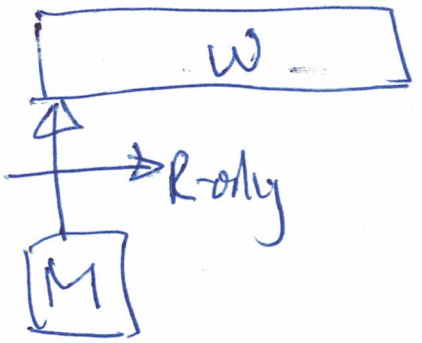
Cpts 317. 10/24

HW Hint. (Key lecture)

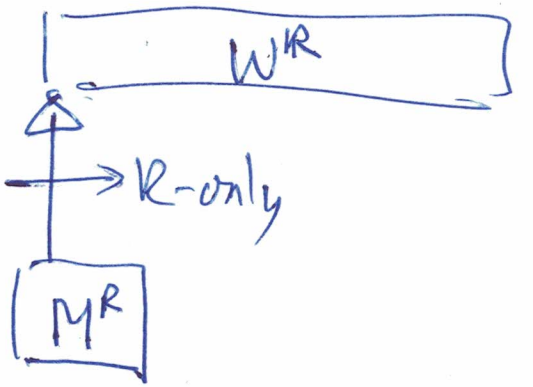
Automata-to-Automata Constructions. (Don't use structural induction).

Prob 3. Show that if L is reg, then so is L^R.

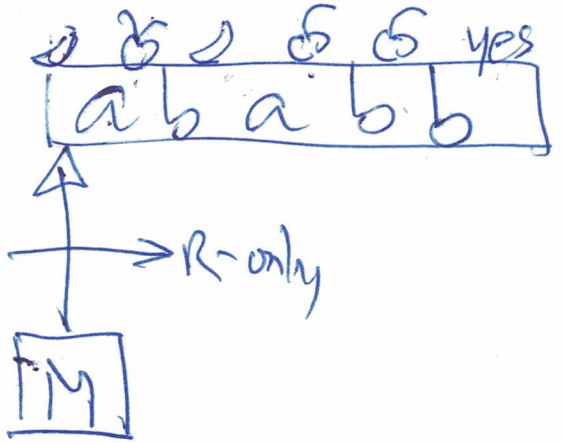
Proof.

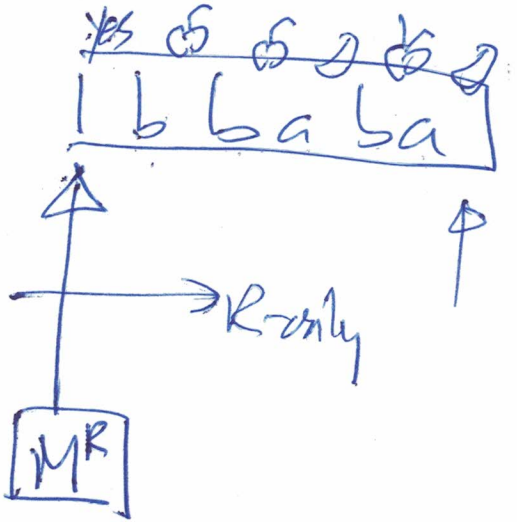


FA M accepts L



M^R accepts L^R





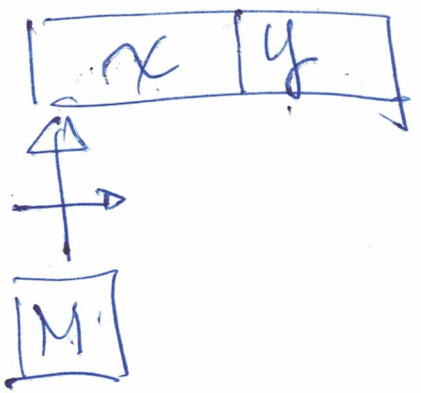
M accepts w iff M^R accepts w^R.

you write it up !!!

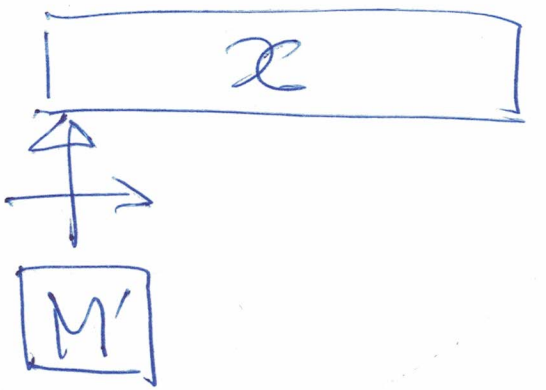
(M^R runs on w^R and simulates M runs on w, reversely).

Prob 4. Show that $Prefix(L) = \{x : \text{for some } y,\ xy \in L\}$ is reg if L is reg.

Proof.



FA M accepts L.



M' accepts $Prefix(L)$.

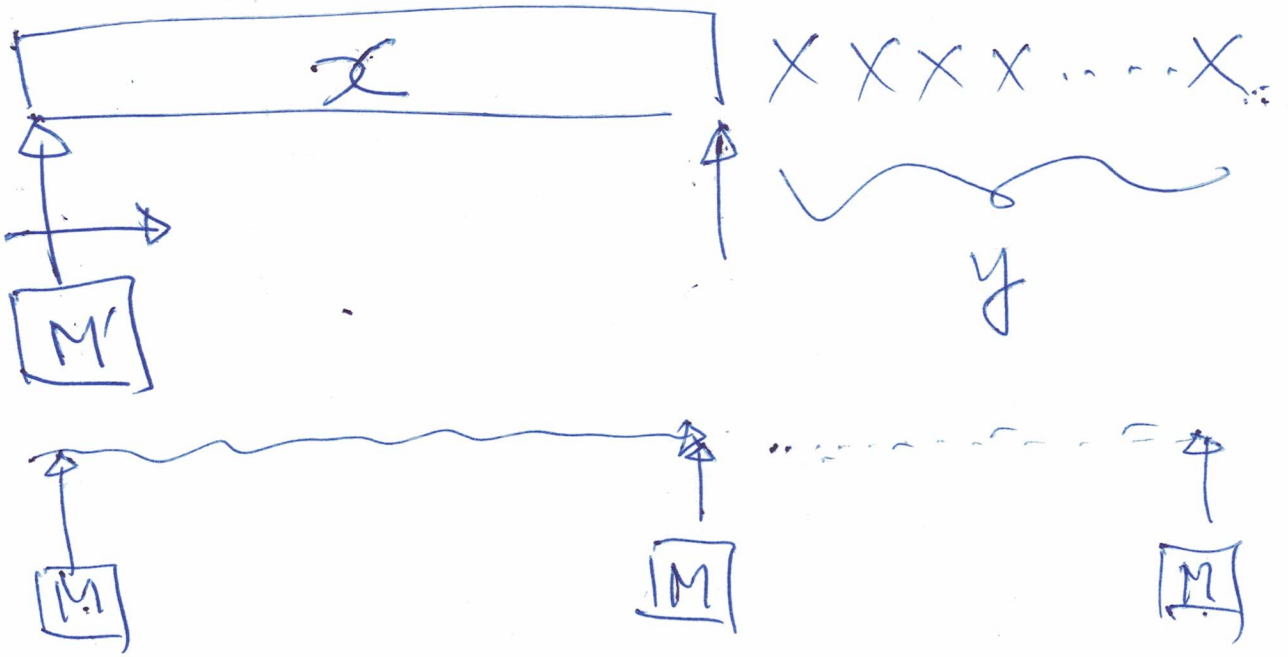
M says yes on xy if M' says yes on x.

for some y

y is <u>not</u> given.

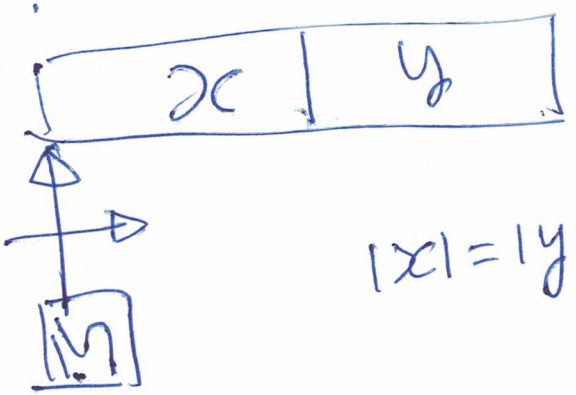
---

you write it up!



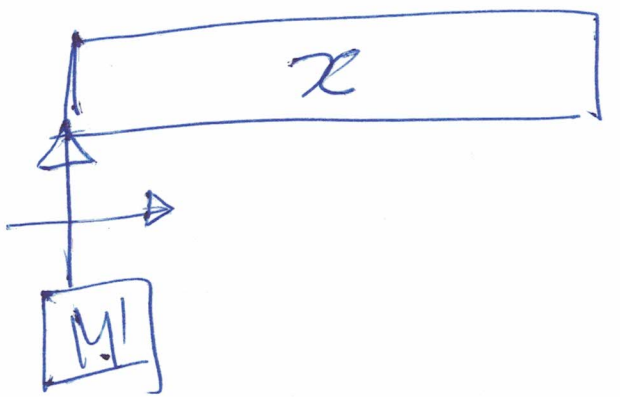
Prob 5. Show that $Half(L) = \{x : \text{for some } y,\ xy \in L \text{ and } |x| = |y|\}$ is reg if L is reg.

Proof.



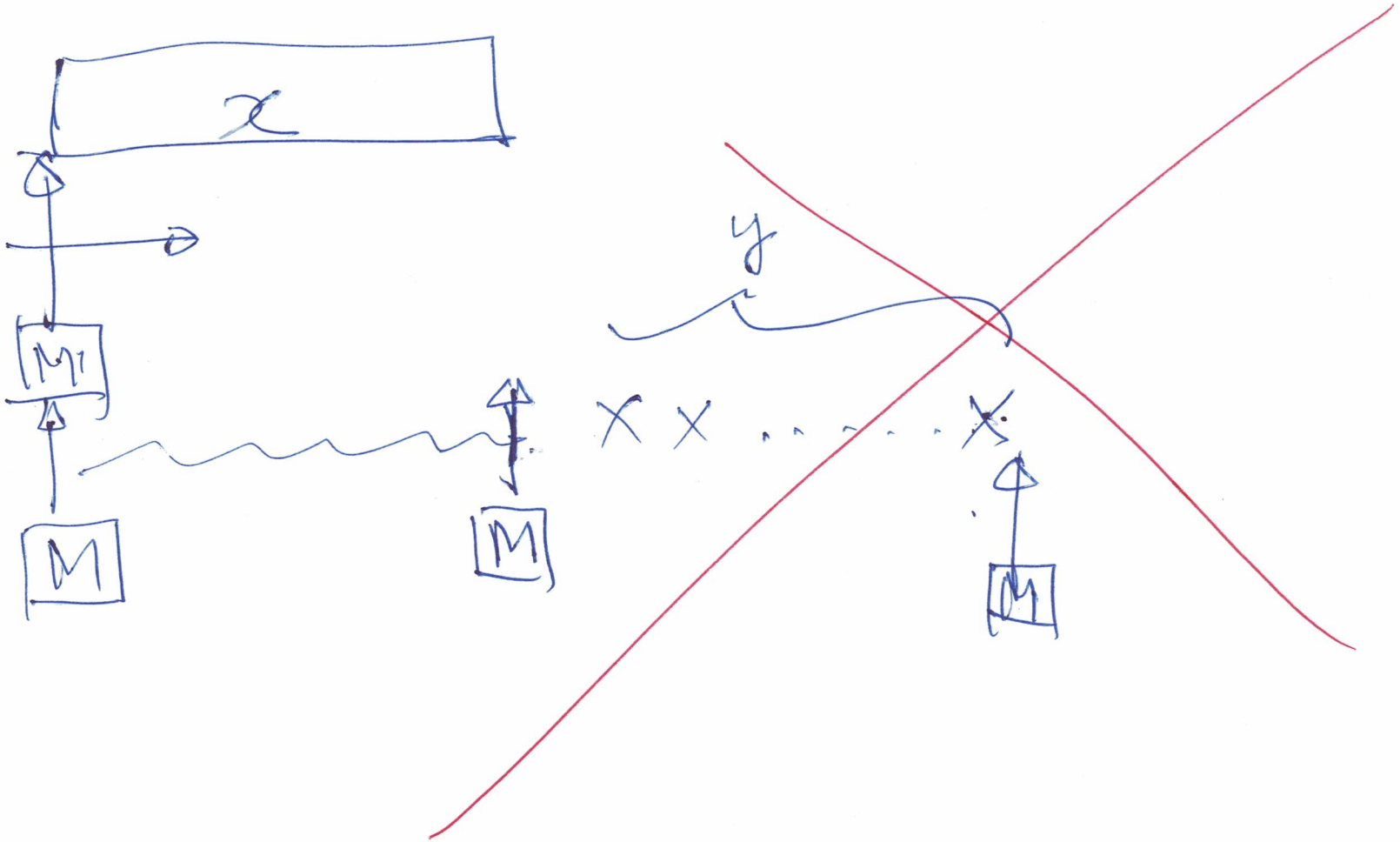
$|x| = |y|$.

M accepts L.

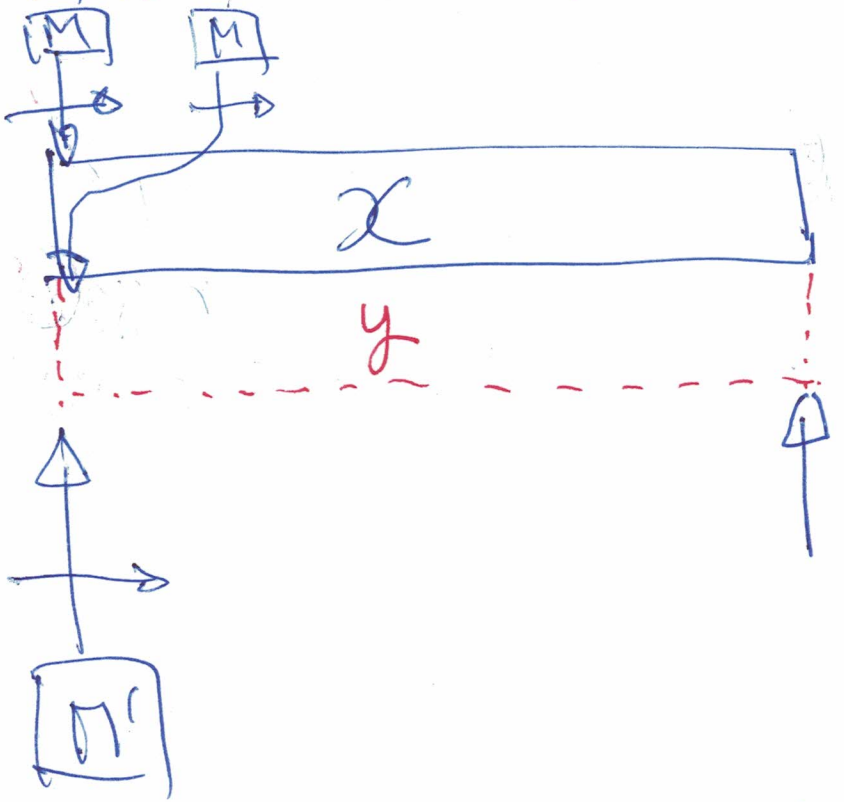


M' accepts $Half(L)$

Wrong solution (construction in previous problem won't work here).



Correct Answer: (you write it up).



(watch the video & write it up).

Prob 5. M', a FA, runs as follows. On input x, it runs two copies M_1 and M_2, in parallel, of M. The first copy runs on input x, while the second copy runs on a guessed y (guessed symbol by symbol as M' reads the input symbols). In particular, the second copy M_2 starts with a guessed state. At the end of input x, M' says yes on x if the first copy M_1 enters the guessed state and the second copy M_2 enters accepting state. Clearly, M' accepts Half(L). ∎

Final Exam Exercise:

Show that $\text{Third}(L) = \{x : \exists y.\ xy \in L$ and $|y| = 2 \cdot |x|\}$ is reg if L is reg.

Prob 2. Show that we have an alg to check:

Given: M_1 & M_2

Contn: $\exists x$ s.t. $(x \in L(M_1)$ & $x \notin L(M_2))$ or $(x \notin L(M_1)$ & $x \in L(M_2))$.

hint: We need only show the emptiness of:

$$\underbrace{(L(M_1) - L(M_2)) \cup (L(M_2) - L(M_1))}_{\text{is regular.}} = \emptyset \,?$$